C.A.N.I.S.

Containment Analysis & Nuclear Intelligence System

DRUM STORE INSPECTION REPORT

CANIS-083203-wh_rigid_walk-de01
Warehouse v3, aisle A (simulated)
2026-11-08 08:32:05 UTC

VERDICT

8 drums need action. 28 showed no problem.

3 aisle sections restricted; 6 not checked. Radiation was not measured on this round, so no part of the aisle can be called low gamma yet.

Action needed 8 · Check again 0 · No problem found 28

DRUMS CHECKED	ACTION NEEDED	CHECK AGAIN	NO PROBLEM FOUND
36 / 36 verdict reached	8 red	0 amber	28 green
PEAK GAMMA	**AISLE SECTIONS**	**ROBOT DOSE**	**MISSION TIME**
n/m not measured this round	9 3 restricted · 6 not checked	n/m not measured this round	7 min 05 s simulated time

WHAT TO DO NOW — MOST URGENT FIRST

	DRUM	RESULT	SEEN	DO THIS
1	**A-L-01-T1**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).
2	**A-L-03-T1**	ACTION NEEDED	Bulging	Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).
3	**A-L-03-T3**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).
4	**A-L-11-T2**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).
5	**A-R-01-T1**	ACTION NEEDED	Bulging	Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).
6	**A-R-02-T2**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).
7	**A-R-04-T2**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).
8	**A-R-09-T2**	ACTION NEEDED	Rust	Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

THIS INSPECTION

Store	Warehouse v3, aisle A (simulated)	Report	CANIS-083203-wh_rigid_walk-de01
Started	2026-11-08 08:32:05 UTC	Mission time	7 min 05 s
Inspected by	C.A.N.I.S. robot (Spot)	Drums in scope	36
Report software	aa9a142	Status	Practice run, simulated store

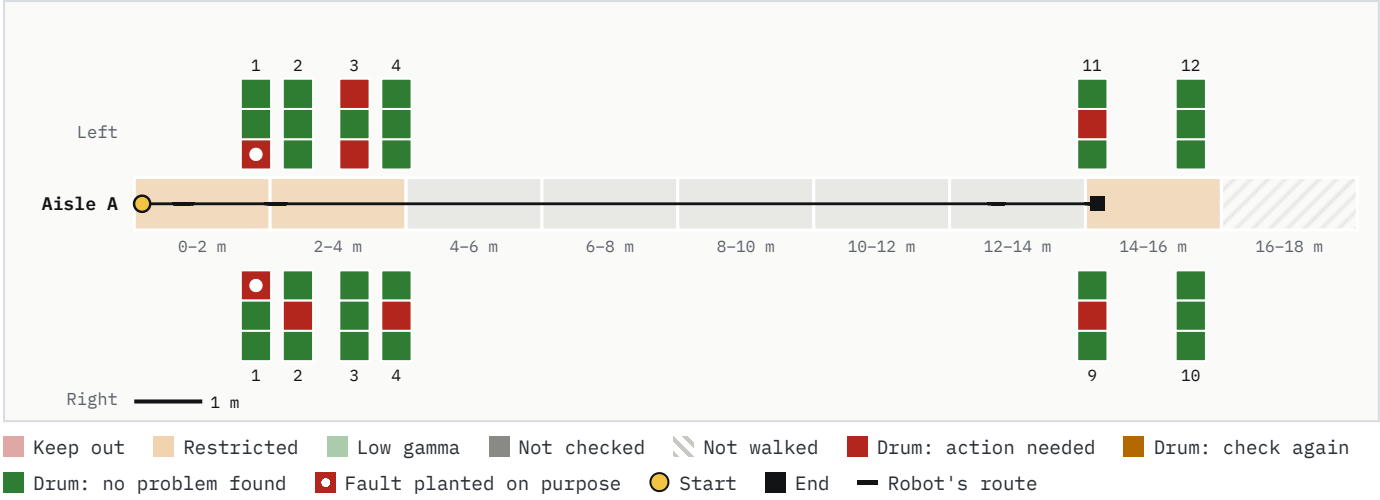
Not measured on this round: radiation, drum surface temperature, hydrogen, surface wipes (swabs). Tables print “not measured”; no value is estimated.

Planted on purpose so the robot’s response shows early: rust on A-L-01-T1, a bulge on A-R-01-T1.

01 Aisle survey

9 SECTIONS · 2 M EACH

Each aisle is split into 2-metre sections measured from where the robot started, and each section takes the worst rating that applies. A section is only rated low gamma if radiation was measured there.



Keep out · Restricted · Low gamma · Not checked · Not walked · Drum: action needed · Drum: check again · Drum: no problem found · Fault planted on purpose · Start · End · Robot's route

Racks are drawn folded out flat: the shelf nearest the aisle is the bottom one. Numbers are stack numbers. Only distance along the aisle was available for the route, so it is drawn on the centre line.

AISLE	SECTION	RATING	SAMPLES	WHY
A	0–2 m	RESTRICTED	108	Drum needing action here: A-L-01-T1, A-R-01-T1. Radiation was not measured, so this may need to be stricter.
A	2–4 m	RESTRICTED	90	Drum needing action here: A-L-03-T1, A-L-03-T3, A-R-02-T2, A-R-04-T2. Radiation was not measured, so this may need to be stricter.
A	14–16 m	RESTRICTED	48	Drum needing action here: A-L-11-T2, A-R-09-T2. Radiation was not measured, so this may need to be stricter.
A	4–6 m	NOT CHECKED	1	The robot walked here, but radiation was not measured.
A	6–8 m	NOT CHECKED	1	The robot walked here, but radiation was not measured.
A	8–10 m	NOT CHECKED	1	The robot walked here, but radiation was not measured.
A	10–12 m	NOT CHECKED	1	The robot walked here, but radiation was not measured.
A	12–14 m	NOT CHECKED	65	The robot walked here, but radiation was not measured.
A	16–18 m	NOT CHECKED	0	The robot did not go here this time.

Samples = telemetry records logged in the section. No radiation or hydrogen readings were logged on this round, so those columns are left out.

02 Every drum at a glance

36 DRUMS · MOST URGENT FIRST

DRUM	WASTE	WHERE	RESULT	WHAT WAS SEEN	HOW SURE	FAULT VIEWS
A-L-01-T1 PLANTED	LLW	Left 1 · bottom shelf	ACTION NEEDED	Rust	High 96%	3/3
A-L-03-T1	LLW	Left 3 · bottom shelf	ACTION NEEDED	Bulging	High 97%	3/3
A-L-03-T3	LLW	Left 3 · top shelf	ACTION NEEDED	Rust	High 83%	7/8
A-L-11-T2	ILW	Left 11 · middle shelf	ACTION NEEDED	Rust	High 95%	3/3
A-R-01-T1 PLANTED	LLW	Right 1 · bottom shelf	ACTION NEEDED	Bulging	High 96%	3/3
A-R-02-T2	LLW	Right 2 · middle shelf	ACTION NEEDED	Rust	High 97%	3/3
A-R-04-T2	LLW	Right 4 · middle shelf	ACTION NEEDED	Rust	High 97%	3/3
A-R-09-T2	ILW	Right 9 · middle shelf	ACTION NEEDED	Rust	High 90%	3/3
A-L-01-T2	LLW	Left 1 · middle shelf	NO PROBLEM FOUND	Nothing found	High 97%	0/3
A-L-01-T3	LLW	Left 1 · top shelf	NO PROBLEM FOUND	Nothing found	High 95%	0/8
A-L-02-T1	LLW	Left 2 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 97%	0/3
A-L-02-T2	LLW	Left 2 · middle shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-L-02-T3	LLW	Left 2 · top shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/8
A-L-03-T2	LLW	Left 3 · middle shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-L-04-T1	LLW	Left 4 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-L-04-T2	LLW	Left 4 · middle shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-L-04-T3	LLW	Left 4 · top shelf	NO PROBLEM FOUND	Nothing found	High 95%	0/8
A-L-11-T1	ILW	Left 11 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 94%	0/3
A-L-11-T3	ILW	Left 11 · top shelf	NO PROBLEM FOUND	Nothing found	High 93%	0/8
A-L-12-T1	ILW	Left 12 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 93%	0/3
A-L-12-T2	ILW	Left 12 · middle shelf	NO PROBLEM FOUND	Nothing found	High 73%	0/3
A-L-12-T3	ILW	Left 12 · top shelf	NO PROBLEM FOUND	Nothing found	High 93%	0/8
A-R-01-T2	LLW	Right 1 · middle shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-R-01-T3	LLW	Right 1 · top shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/8
A-R-02-T1	LLW	Right 2 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-R-02-T3	LLW	Right 2 · top shelf	NO PROBLEM FOUND	Nothing found	High 87%	0/8
A-R-03-T1	LLW	Right 3 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-R-03-T2	LLW	Right 3 · middle shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-R-03-T3	LLW	Right 3 · top shelf	NO PROBLEM FOUND	Nothing found	High 95%	0/8
A-R-04-T1	LLW	Right 4 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 95%	0/3
A-R-04-T3	LLW	Right 4 · top shelf	NO PROBLEM FOUND	Nothing found	High 94%	0/8
A-R-09-T1	ILW	Right 9 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-R-09-T3	ILW	Right 9 · top shelf	NO PROBLEM FOUND	Nothing found	High 94%	0/8
A-R-10-T1	ILW	Right 10 · bottom shelf	NO PROBLEM FOUND	Nothing found	High 96%	0/3
A-R-10-T2	ILW	Right 10 · middle shelf	NO PROBLEM FOUND	Nothing found	High 91%	0/3
A-R-10-T3	ILW	Right 10 · top shelf	NO PROBLEM FOUND	Nothing found	High 92%	0/8

How sure = the detector's confidence in what it reported: High ≥ 70%, Medium ≥ 50%, Low below. Fault views = camera views that showed damage, out of all views taken. Drum names read aisle, side (L / R), stack, shelf (T1 = bottom). LLW = low-level waste (yellow drums), ILW = intermediate-level waste (blue drums). This is the first report for these drums, so there is no earlier inspection to compare with.

03 Drums that need attention

8 DRUMS

A-L-01-T1 Aisle A, left side, stack 1, bottom shelf · Low-level waste

ACTION NEEDED



robot camera view · 00:10.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Rust
How sure	High (96%)
Views	3 of 3 showed damage
Why this rating	Rust confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
Test note	This fault (rust) was planted on purpose for the demonstration.
What the robot did	00:10 Rust seen in 3 of 3 views (how sure: 96%). Action needed.

WHAT TO DO

Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

A-L-03-T1 Aisle A, left side, stack 3, bottom shelf · Low-level waste

ACTION NEEDED



robot camera view · 02:18.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Bulging
How sure	High (97%)
Views	3 of 3 showed damage
Why this rating	Bulging confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
What the robot did	02:18 Bulging seen in 3 of 3 views (how sure: 97%). Action needed.

WHAT TO DO

Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).

A-L-03-T3 Aisle A, left side, stack 3, top shelf · Low-level waste

ACTION NEEDED



robot camera view · 03:34.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Rust
How sure	High (83%)
Views	7 of 8 showed damage
Why this rating	Rust confirmed in 7 of 8 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
What the robot did	02:28 **Asked for a closer look.** "Drum A-L-03-T3 shows signs of rust and discoloration, requiring a…" 03:34 Rust seen in 7 of 8 views (how sure: 83%). Action needed.

WHAT TO DO

Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

A-L-11-T2 Aisle A, left side, stack 11, middle shelf · Intermediate-level waste

ACTION NEEDED



robot camera view · 04:56.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Rust
How sure	High (95%)
Views	3 of 3 showed damage
Why this rating	Rust confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
What the robot did	04:55 Rust seen in 3 of 3 views (how sure: 95%). Action needed.

WHAT TO DO

Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

A-R-01-T1 Aisle A, right side, stack 1, bottom shelf · Low-level waste

ACTION NEEDED



robot camera view · 00:46.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Bulging
How sure	High (96%)
Views	3 of 3 showed damage
Why this rating	Bulging confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
Test note	This fault (a bulge) was planted on purpose for the demonstration.
What the robot did	00:46 Bulging seen in 3 of 3 views (how sure: 96%). Action needed.

WHAT TO DO

Limit access to this aisle. Ask an engineer to assess the drum for pressure build-up, and measure its surface temperature (not measured in this inspection).

A-R-02-T2 Aisle A, right side, stack 2, middle shelf · Low-level waste

ACTION NEEDED



robot camera view · 01:09.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

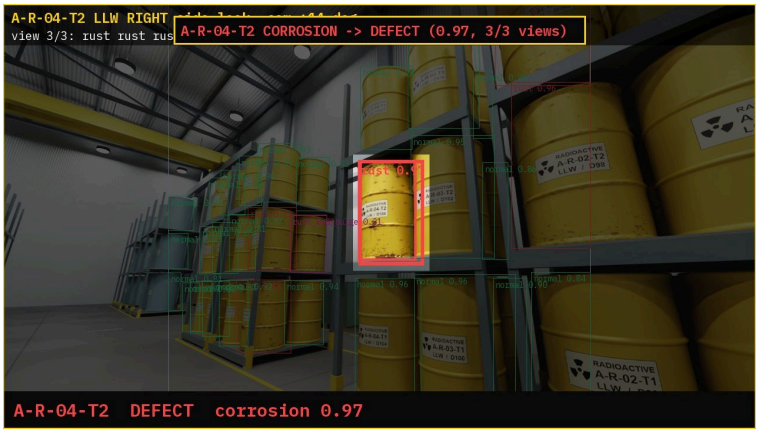
What was seen	Rust
How sure	High (97%)
Views	3 of 3 showed damage
Why this rating	Rust confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
What the robot did	01:08 Rust seen in 3 of 3 views (how sure: 97%). Action needed.

WHAT TO DO

Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

A-R-04-T2 Aisle A, right side, stack 4, middle shelf · Low-level waste

ACTION NEEDED



robot camera view · 03:19.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Rust
How sure	High (97%)
Views	3 of 3 showed damage
Why this rating	Rust confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
What the robot did	03:18 Rust seen in 3 of 3 views (how sure: 97%). Action needed.

WHAT TO DO

Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

A-R-09-T2 Aisle A, right side, stack 9, middle shelf · Intermediate-level waste

ACTION NEEDED



robot camera view · 05:17.0 into the run. Overlay, boxes and labels are the robot's own, as recorded.

What was seen	Rust
How sure	High (90%)
Views	3 of 3 showed damage
Why this rating	Rust confirmed in 3 of 3 views.
Surface wipe	not taken
Not measured	gamma at drum, surface temp., hydrogen
What the robot did	05:16 Rust seen in 3 of 3 views (how sure: 90%). Action needed.

WHAT TO DO

Ask an engineer to assess how far the rust has gone. Take a surface wipe if none was taken. Consider placing the drum in an overpack (a larger outer drum).

05 What this report cannot tell you

- The robot reached a verdict on all 36 drums it was sent to look at.
- **5 sections** (A 4–6 m, A 6–8 m, A 8–10 m, A 10–12 m, A 12–14 m): walked, but radiation was not measured.
- **1 section** (A 16–18 m): not walked on this round.
- **Radiation**: not measured anywhere on this round.
- **Drum surface temperature**: not measured anywhere on this round.
- **Hydrogen**: not measured anywhere on this round.
- **Surface wipes (swabs)**: not measured anywhere on this round.

How accurate was the robot?

This was a practice run in a simulated store, so we know which drums were really damaged. The robot was not told; the comparison is made only for this report.

DAMAGED DRUMS FOUND	MISSED	FALSE ALARMS
8 / 11 incl. 2 planted	3 the more serious error	0 each costs one extra check

MISSED DRUM	SIMULATED DAMAGE (0–1)	WHERE
A-L-12-T1	bulge 0.95	Aisle A, left side, stack 12, bottom shelf
A-R-02-T3	rust 0.54, bulge 0.49	Aisle A, right side, stack 2, top shelf
A-R-09-T1	bulge 0.17	Aisle A, right side, stack 9, bottom shelf

DRUM COLOUR	DRUMS	CORRECT	ACCURACY	95% RANGE	DAMAGED FOUND	FALSE ALARMS
Yellow (low-level)	24	23	**95.8%**	80–99%	6 of 7	0
Blue (intermediate-level)	12	10	**83.3%**	55–95%	2 of 4	0

A drum counts as correct if it was damaged and flagged, or undamaged and cleared. Yellow excluding the 2 planted drums: 95.5% (21 of 22). With this few drums the 95% ranges are wide, so small differences between colours should not be over-read.

How the ratings were made

PART	DOES	NEVER DOES
Detector	Draws a box and a class (normal / rust / bulge) on each camera view.	Decide a rating.
Rules	Turn views and readings into drum results and aisle ratings, the same way every time.	Read free text.
Cosmos Reason 2	Chooses from a fixed list of options: accept the rules' result, ask for a closer look, or mark a drum for a person. Its reasons are quoted in the record.	Clear a drum, confirm damage, or change any rating, zone or number.
Report	Lays out what the run logged.	Estimate a value that was not logged.

On this round Cosmos Reason 2 made 29 reviews: 17 accepted the rules' result, 12 asked for a closer look.

Limits used

RATING	LEVEL	SOURCE
Restricted area	gamma > 7.5 µSv/h	HSE, Work with ionising radiation (L121), regulation 17. Used as an instantaneous robot rule, not a legal controlled-area assessment.
Keep out	gamma > 100 µSv/h	HSE L121 guidance on high instantaneous dose rates.
Surface wipe	> 0.4 / 4 Bq/cm²	IAEA SSR-6 Rev. 2 (2025), para 508. Transport limits borrowed as escalation values.

RATING	LEVEL	SOURCE
Warm drum	≥ 5 °C above air	Team demo rule. Not an operational limit.
Hydrogen	no level set	No cited alarm level yet. Readings are printed; hydrogen changes no rating.

How to read the labels

ACTION NEEDED	Damage confirmed from more than one view. An engineer should assess the drum.	KEEP OUT	Highest-priority simulation flag.
CHECK AGAIN	Possible damage, or the robot could not confirm the drum.	RESTRICTED	Raised gamma, or a damaged or unresolved drum in the section.
NO PROBLEM FOUND	No damage seen in this inspection.	LOW GAMMA	Gamma measured and below 7.5 µSv/h.
		NOT CHECKED	No radiation reading here: treat as unknown.

Practice run in a computer-simulated store (NVIDIA Isaac Sim). The detector was trained on synthetic images; rust extent is judged from photos, not measured. Only a surface wipe can show contamination. The structure follows the themes of the NDA's Guidance on Monitoring of Waste Packages During Storage (WPS/640).

Video evidence

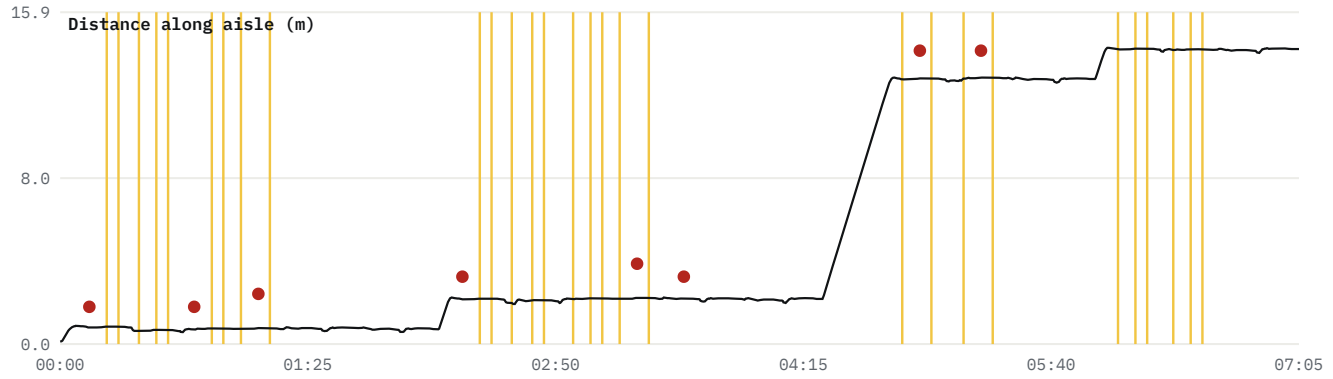
The whole round was recorded on video, about 7 minutes long. A 58-second showcase cut shows the moments below in real time. Everything between them is sped up about 20 times, and the clock on screen always shows true mission time.

IN THE CUT	WHAT IT SHOWS
0:00–0:13	Rust found on A-L-01-T1, then the drum above it (A-L-01-T2) cleared as normal
0:19–0:23	A-L-03-T3 on the top shelf: Cosmos Reason 2 asks for a closer look
0:26–0:37	The robot comes back to A-L-03-T3 and confirms rust in 7 of 8 views
0:47–0:58	The last drum, then the end-of-round summary

A Readings over time

MISSION CLOCK MM:SS

No sensor readings were logged on this round, so this shows how far down the aisle the robot was. Flat stretches are where it stopped to look at drums; dots mark drums found to need action; yellow lines mark Cosmos Reason 2 decisions.



B Step-by-step record

65 LOGGED EVENTS

What the robot decided and why, in order. Cosmos Reason 2 reasons are quoted exactly as logged (the log shortens long ones); they choose the next step and never set a rating.

TIME	DRUM	SOURCE	WHAT HAPPENED	MS
00:10.0	A-L-01-T1	CANIS RULES	Rust seen in 3 of 3 views (how sure: 96%). Action needed.	
00:16.0	A-L-01-T2	COSMOS REASON 2	**Accepted the rules' result.** "Drum appears intact with no visible damage or discoloration, and..."	
00:16.0	A-L-01-T2	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
00:20.0	A-L-01-T3	COSMOS REASON 2	**Asked for a closer look.** "The drum appears intact with no visible damage or discoloration, but..."	
00:27.0	A-L-02-T1	COSMOS REASON 2	**Accepted the rules' result.** "Drum appears intact with no visible damage or discoloration, and..."	
00:27.0	A-L-02-T1	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
00:33.0	A-L-02-T2	COSMOS REASON 2	**Accepted the rules' result.** "Drum A-L-02-T2 shows no visible damage, and the detector confirms it..."	
00:33.0	A-L-02-T2	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
00:37.0	A-L-02-T3	COSMOS REASON 2	**Asked for a closer look.** "The drum shows no visible damage or discoloration, but the rule..."	
00:46.0	A-R-01-T1	CANIS RULES	Bulging seen in 3 of 3 views (how sure: 96%). Action needed.	
00:52.0	A-R-01-T2	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage or discoloration, and..."	
00:52.0	A-R-01-T2	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
00:56.0	A-R-01-T3	COSMOS REASON 2	**Asked for a closer look.** "The drum appears intact with no visible damage or discoloration, but..."	
01:02.0	A-R-02-T1	COSMOS REASON 2	**Accepted the rules' result.** "Drum appears intact with no visible damage or discoloration, and..."	
01:02.0	A-R-02-T1	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
01:08.0	A-R-02-T2	CANIS RULES	Rust seen in 3 of 3 views (how sure: 97%). Action needed.	
01:12.0	A-R-02-T3	COSMOS REASON 2	**Asked for a closer look.** "Drum A-R-02-T3 shows slight rust and discoloration, requiring closer..."	

TIME	DRUM	SOURCE	WHAT HAPPENED	MS
01:24.0	**A-L-01-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
01:38.0	**A-R-01-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
01:51.0	**A-L-02-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
02:04.0	**A-R-02-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
02:18.0	**A-L-03-T1**	CANIS RULES	Bulging seen in 3 of 3 views (how sure: 97%). Action needed.	
02:24.0	**A-L-03-T2**	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage or discoloration, and..."	
02:24.0	**A-L-03-T2**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
02:28.0	**A-L-03-T3**	COSMOS REASON 2	**Asked for a closer look.** "Drum A-L-03-T3 shows signs of rust and discoloration, requiring a..."	
02:35.0	**A-L-04-T1**	COSMOS REASON 2	**Accepted the rules' result.** "Drum appears intact with no visible damage or discoloration, and..."	
02:35.0	**A-L-04-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
02:42.0	**A-L-04-T2**	COSMOS REASON 2	**Accepted the rules' result.** "Drum A-L-04-T2 shows no visible damage, and the detector confirms it..."	
02:42.0	**A-L-04-T2**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
02:46.0	**A-L-04-T3**	COSMOS REASON 2	**Asked for a closer look.** "The drum shows no visible damage or discoloration, but the rule..."	
02:56.0	**A-R-03-T1**	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage, and the detector..."	
02:56.0	**A-R-03-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
03:02.0	**A-R-03-T2**	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage, and the detector..."	
03:02.0	**A-R-03-T2**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
03:06.0	**A-R-03-T3**	COSMOS REASON 2	**Asked for a closer look.** "The drum appears intact with no visible damage or discoloration, but..."	
03:12.0	**A-R-04-T1**	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage or discoloration, and..."	
03:12.0	**A-R-04-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
03:18.0	**A-R-04-T2**	CANIS RULES	Rust seen in 3 of 3 views (how sure: 97%). Action needed.	
03:22.0	**A-R-04-T3**	COSMOS REASON 2	**Asked for a closer look.** "Drum A-R-04-T3 shows rust and discoloration, requiring a closer..."	
03:34.0	**A-L-03-T3**	CANIS RULES	Rust seen in 7 of 8 views (how sure: 83%). Action needed.	
03:47.0	**A-R-03-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
04:02.0	**A-L-04-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
04:15.0	**A-R-04-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
04:49.0	**A-L-11-T1**	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage, and the detector..."	
04:49.0	**A-L-11-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
04:55.0	**A-L-11-T2**	CANIS RULES	Rust seen in 3 of 3 views (how sure: 95%). Action needed.	
04:59.0	**A-L-11-T3**	COSMOS REASON 2	**Asked for a closer look.** "The drum appears intact with no visible damage or discoloration, but..."	
05:10.0	**A-R-09-T1**	COSMOS REASON 2	**Accepted the rules' result.** "The drum appears intact with no visible damage, and the detector..."	
05:10.0	**A-R-09-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	

TIME	DRUM	SOURCE	WHAT HAPPENED	MS
05:16.0	**A-R-09-T2**	CANIS RULES	Rust seen in 3 of 3 views (how sure: 90%). Action needed.	
05:20.0	**A-R-09-T3**	COSMOS REASON 2	**Asked for a closer look.** “Rust and discoloration on the drum indicate potential damage…”	
05:35.0	**A-L-11-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
05:49.0	**A-R-09-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
06:03.0	**A-L-12-T1**	COSMOS REASON 2	**Accepted the rules’ result.** “The drum appears intact with no visible damage, and the detector…”	
06:03.0	**A-L-12-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
06:09.0	**A-L-12-T2**	COSMOS REASON 2	**Accepted the rules’ result.** “The drum appears intact with no visible damage or discoloration, and…”	
06:09.0	**A-L-12-T2**	CANIS RULES	No damage seen (2 of 3 views looked normal). No problem found.	
06:13.0	**A-L-12-T3**	COSMOS REASON 2	**Asked for a closer look.** “The drum appears intact with no visible damage or discoloration, but…”	
06:22.0	**A-R-10-T1**	COSMOS REASON 2	**Accepted the rules’ result.** “The drum appears intact with no visible damage or discoloration, and…”	
06:22.0	**A-R-10-T1**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
06:28.0	**A-R-10-T2**	COSMOS REASON 2	**Accepted the rules’ result.** “The drum appears intact with no visible damage, and the detector…”	
06:28.0	**A-R-10-T2**	CANIS RULES	No damage seen (3 of 3 views looked normal). No problem found.	
06:32.0	**A-R-10-T3**	COSMOS REASON 2	**Asked for a closer look.** “The drum appears intact with no visible damage or discoloration, but…”	
06:45.0	**A-L-12-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	
06:58.0	**A-R-10-T3**	CANIS RULES	No damage seen (8 of 8 views looked normal). No problem found.	

C Technical details

FOR ENGINEERS AND AUDITORS

Report	CANIS-083203-wh_rigid_walk-de01
Run folder	takes/083203/run
Input format	framework
Report software	aa9a142
Detector	YOLO26s whole-barrel conditions, run compare_colors_20261108T053740Z_continued_yolo26s
Detector SHA-256	ba4c80bcb99f6152d8ee0a6523151c4640ad67d727431b9ec846235dcf895b6a
Decision rule	Side look, 3 views: fault (≥0.50) in ≥2 views →defect; normal in ≥2 and no fault → OK; otherwise, and always for tier 3, a closer look (8 views in total): fault in ≥4, or ≥3 and more than normal →defect; normal in ≥5 and fault ≤1 →OK; otherwise suspect.
Detector calls	168 made, 0 failed
Planted faults	A-L-01-T1 rust; A-R-01-T1 a bulge
Ground truth	simulation inventory seed 42 (scored by wh_score_color.py)
Run ID	083203-wh_rigid_walk-de01
Walk software	217e679 plus uncommitted walk changes
Walk script	scripts/hot/full_hall/wh_rigid_walk.py with wh_rigid_full_cosmos.py
Cosmos reviews	NVIDIA Cosmos Reason 2 8B: 29 reviews, 29 valid, 17 accepted, 12 closer looks, 0 marked suspect; slowest reply 2.2 s. Options offered: accept or closer look when the rules say OK; closer look or mark suspect when unclear or top shelf. Never offered when the rules confirm damage.
Planted fault severity	A-L-01-T1 rust 0.85; A-R-01-T1 bulge 0.85. All other faults: inventory seed 42.
Logged channels	Robot pose, camera tilt, detector boxes, decisions, Cosmos reviews. No dose rate, temperature, gas or swab.
Simulation	NVIDIA Isaac Sim, warehouse-v3
Report source	Rebuilt from final_canis_showcase_report.pdf: tables and text by word position, route from its vector chart, stack positions from its vector plan, frames as embedded.
Status names	Action needed = Confirmed defect · Check again = Suspect or Couldn’t inspect · No problem found = OK

Evidence frames

DRUM	VIEW	TIME	FRAME CLASS	CONF.	PITCH	SOURCE
A-L-01-T1	robot camera view	00:10.0				A-L-01-T1_0m10s.jpeg
A-L-03-T1	robot camera view	02:18.0				A-L-03-T1_2m18s.jpeg
A-L-03-T3	robot camera view	03:34.0				A-L-03-T3_3m34s.jpeg
A-L-11-T2	robot camera view	04:56.0				A-L-11-T2_4m56s.jpeg
A-R-01-T1	robot camera view	00:46.0				A-R-01-T1_0m46s.jpeg
A-R-02-T2	robot camera view	01:09.0				A-R-02-T2_1m09s.jpeg
A-R-04-T2	robot camera view	03:19.0				A-R-04-T2_3m19s.jpeg
A-R-09-T2	robot camera view	05:17.0				A-R-09-T2_5m17s.jpeg